Traffic Simulation User Guide

Real-Time Traffic Simulation System

User Guide - Milestone 2

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Project Team: SUMO Traffic Simulation
Course: Object-Oriented Programming in Java

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1. Introduction

1.1 What is This Application?

This is a traffic simulation program that connects to SUMO (Simulation of Urban MObility) to help visualize and control traffic. You can:

- See traffic moving on a road network
- Add vehicles to the simulation
- Control traffic lights
- Export data for analysis

1.2 What You Need

Software Required: - Java JDK 17 or higher - SUMO 1.24 or 1.25 (traffic simulator) - Eclipse IDE (recommended)

Your Computer: - Windows, Mac, or Linux - 4 GB RAM minimum - Any modern processor

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2. Quick Start Guide

Step 1: Setup (One-Time)

Install SUMO: 1. Download from https://sumo.dlr.de/docs/Downloads.php 2. Install to default location 3. Remember where it installed!

Set SUMO_HOME:

Windows:

```
setx SUMO_HOME "C:\Program Files\Eclipse\Sumo"
```

Mac/Linux:

```
export SUMO_HOME=/usr/local/share/sumo
echo 'export SUMO_HOME=/usr/local/share/sumo' >> ~/.bashrc
```

IMPORTANT: Restart your computer after setting SUMO_HOME!

Step 2: Configure the Project

1. Open Eclipse

2. Import project: File → Import → Existing Projects

3. **CRITICAL STEP**: Edit `SimulationManager.java` line 40

```java
        // Change this path to YOUR network file:
        private static final String CONFIG_FILE = "/path/to/your/
network.sumocfg";
```

Use a real SUMO network file path (absolute path recommended)

Step 3: First Run

1. Right-click `MainApp.java` → Run As → Java Application
2. Window opens (might take 10 seconds)
3. Click **“Start Simulation”** button
4. Wait 5-10 seconds for map to load
5. Roads appear as gray lines - you’re ready!

Step 4: Add Your First Vehicle

1. Click **“Add Vehicle”** button
2. Fill in:
    - Vehicle ID: `car1`
    - Route: Pick any from dropdown
    - Type: `Car`
    - Color: `Red`
3. Click **“Add”**
4. Red dot appears and starts moving!

That’s it! You’re running the simulation.

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3. User Interface

3.1 Main Window

SUMO Traffic Simulator Interface

Figure 1: Main application window showing the map area (left) with road network and traffic lights, and the action panel (right) with control buttons.

3.2 What You See

Map (center): - Gray lines = roads - Colored dots = vehicles - Small circles = traffic lights (red/green/yellow)

Buttons (right): - All the controls you need

Bottom: - Start/Stop buttons

4. Using the Application

4.1 Adding Vehicles

Single Vehicle

1. Click **“Add Vehicle”**
2. Enter details:
    - **Vehicle ID**: Must be unique (e.g., car1, car2, taxi1)
    - **Route ID**: Where the vehicle will drive
    - **Type**: Car, Bike, or Taxi
    - **Color**: Visual only
3. Click **“Add”**

Tips: - Routes must exist in your SUMO network - Use simple IDs like: car1, car2, car3

Random Vehicle

- Click **“Add Random Vehicle”** - that’s it!
- Creates vehicle with random route and color
- Good for quick testing

Many Vehicles (Flow)

1. Click **“Trigger Flow”**
2. Enter:
    - Base ID (e.g., flow1)
    - Route
    - Type and Color
    - **#Vehicle**: How many (try 10-20 first)
3. Click **“Add”**
4. All vehicles appear at once!

Example: Enter 20 for #Vehicle → creates 20 cars instantly

Stress Test

1. Click **“Stress Test”**
2. Enter number (e.g., 50)
3. Click **“Start test”**
4. Vehicles added slowly (one every 0.2 seconds)

When to use what: - Single Vehicle: Testing specific scenarios - Random: Quick population - Flow: Rush hour simulation - Stress Test: Performance testing

4.2 Controlling Traffic Lights

View and Control: 1. Click **“Change Light”** button 2. Select traffic light from dropdown 3. See current phase info 4. Click action: - **Switch to RED**: Stop traffic - **Switch to GREEN**: Allow traffic - **Switch to YELLOW**: Transition

What the lights do: - Red = Stop - Green = Go - Yellow = Caution

Automatic Control: The system has built-in adaptive control: - Detects congested lanes - Automatically switches lights to help - Runs every simulation step

You can override manually, but adaptive control takes back over after next phase change.

4.3 Filtering Vehicles

To see only certain vehicles: 1. Click “Filter Vehicles” checkbox 2. Choose filters: - **Speed Range**: 0-5 (stopped), 5-10 (slow), 10-15 (normal), etc. - **Color**: Red, Blue, Green, etc. 3. Click **“Apply Filters”** 4. Map shows only matching vehicles

Uncheck the box to see all vehicles again.

Common uses: - Speed 0-5: Find traffic jams - Specific color: Track a group of vehicles

4.4 Map Controls

Zoom: - Mouse wheel up = zoom in - Mouse wheel down = zoom out

Pan: - Click and drag the map

Tips: - Zoom out first to see whole network - Then zoom in on areas of interest

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5. Data Export

5.1 What Gets Exported

Two CSV files are created automatically:

1. vehicle_report.csv - Contains: Vehicle positions and speeds - Updated: Every simulation second - Location: Project folder

2. trafficlight_report.csv - Contains: Traffic light states - Updated: Every simulation second - Location: Project folder

5.2 File Format

vehicle_report.csv:

```
Time;VehicleID;Speed_ms;X_Coord;Y_Coord
12.5;car1;13.25;245.18;189.32
12.5;car2;8.42;312.73;220.15
```

Columns: - Time: Simulation time (seconds) - VehicleID: Vehicle name - Speed_ms: Speed in meters/second - X_Coord, Y_Coord: Position

trafficlight_report.csv:

```
Time;LightID;ProgramID;CurrentPhase;Duration
12.5;tl_0;0;2;42.5
```

Columns: - Time: Simulation time - LightID: Traffic light name - ProgramID: Usually 0 - CurrentPhase: Phase number (0, 1, 2…) - Duration: Seconds remaining

5.3 Using the Data

In Excel: 1. Open CSV file 2. Create graphs 3. Calculate average speed 4. Find patterns

Example Python:

```python
import pandas as pd

# Load data
df = pd.read_csv('vehicle_report.csv', sep=';')

# Calculate average speed
avg_speed = df['Speed_ms'].mean()
print(f"Average speed: {avg_speed:.2f} m/s")

# Find stopped vehicles
stopped = df[df['Speed_ms'] < 0.5]
print(f"Stopped vehicles: {len(stopped)}")
```

IMPORTANT: Always click “Close Simulation” button before opening CSV files! This ensures all data is saved.

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6. Troubleshooting

Problem: Application Won’t Start

Error: “SUMO_HOME not set”

Solution: 1. Open command prompt/terminal 2. Check: `echo %SUMO_HOME%` (Windows) or `echo $SUMO_HOME` (Mac/Linux) 3. If empty, set it (see Quick Start Step 1) 4. **Restart computer** after setting 5. Try again

Problem: “Cannot connect to SUMO”

Error on clicking Start Simulation

Solutions: 1. Check the config file path in `SimulationManager.java` line 40 2. Make sure the file exists: - Right-click on path in code - Should open valid .sumocfg file 3. Test SUMO separately: `bash    sumo-gui -c /path/to/your/config.sumocfg` If SUMO GUI opens, path is correct

Problem: Map is Blank

Roads don’t appear after starting

Solutions: 1. Wait 10-30 seconds (large networks are slow) 2. Check console for errors 3. Try zooming out (mouse wheel down) 4. Restart simulation

Problem: “Route doesn’t exist”

When adding vehicle

Solutions: 1. Wait for full initialization (watch console) 2. Use “Add Random Vehicle” instead 3. Check your SUMO network has routes defined

Problem: Vehicles Don’t Appear

Vehicle created but not visible

Solutions: 1. Zoom out completely 2. Check console - vehicle might have arrived immediately 3. Try different route (longer distance) 4. Check simulation is running (console shows time updates)

Problem: CSV Files Empty

Files exist but no data

Solutions: 1. Add vehicles first! 2. Let simulation run (need vehicles moving) 3. **Always click “Close Simulation”** - don’t just close window 4. Check file permissions in project folder

Common Mistakes

1. Wrong CONFIG_FILE path - Most common problem! - Use absolute path like: `C:/Users/YourName/sumo/network.sumocfg` - Use forward slashes `/` even on Windows

2. Forgot SUMO_HOME - Set it once, restart computer, never worry again - Check with: `sumo --version` in terminal

3. Didn't wait for initialization - Starting simulation takes 5-30 seconds - Watch console messages - Don't add vehicles until you see "Simulation cycle starting…"

4. Closed window instead of clicking button - Always use "Close Simulation" button - This saves your CSV files properly

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Appendix A: Quick Reference

Starting Up

```
1. Open Eclipse
2. Run MainApp.java
3. Click "Start Simulation"
4. Wait for map to load
```

Adding Vehicles

```
Single: Click "Add Vehicle" → Fill form
Random: Click "Add Random Vehicle"
Many: Click "Trigger Flow" → Enter count
Test: Click "Stress Test" → Enter count
```

Controls

```
Zoom: Mouse wheel
Pan: Click and drag
Filter: Click checkbox → Choose filters
Lights: Click "Change Light" → Select → Action
```

Files Created

```
vehicle_report.csv - Vehicle data
trafficlight_report.csv - Light data
Both in project root folder
```

If Something Goes Wrong

```
1. Check console for errors
2. Read error message carefully
3. Check troubleshooting section
4. Restart simulation
5. Restart Eclipse if needed
```

Appendix B: SUMO Resources

Get Example Networks: - Look in: `$SUMO_HOME/share/sumo/data/examples` - Try: `hello/hello.sumocfg` for simple network

SUMO Documentation: - Main: https://sumo.dlr.de/docs/ - Tutorials: https://sumo.dlr.de/docs/Tutorials.html

Testing SUMO Works:

```
# Run this in terminal:
sumo --version

# Should show: Eclipse SUMO Version X.X.X
```

Summary

This application helps you: 1. Visualize traffic simulation 2. Add and control vehicles 3. Manage traffic lights 4. Export data for analysis

Remember: - Set SUMO_HOME first time only - Fix CONFIG_FILE path in code - Click “Start Simulation” and wait - Use “Close Simulation” button to save data

Most Common Issues: 1. SUMO_HOME not set → Set it, restart computer 2. Wrong CONFIG_FILE path → Use absolute path 3. Impatient → Wait for initialization!

For Help: - Check console output - Read error messages - Review troubleshooting section

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End of User Guide

Good luck with your simulation!
